6. Project Testing

Project Name: FitBuddy - AI Fitness Plan Generator using Gemini Models

Detailed Project Phase Documentation

1. Testing Overview

Testing verifies that FitBuddy works correctly for valid and invalid inputs and that the AI integration produces a usable response.

Tabulation

Test Type	What is checked
Functional	Features work as expected
Integration	Frontend/backend/Gemini communication
UI	Layout and readability
Error	Invalid input/API failure handling

Coding / Technical Example

```
def test_plan(plan):
    assert plan is not None
    assert len(plan) > 0
    print("PASS")

test_plan("Sample FitBuddy plan")
```

Picture / Diagram

```
INPUT TEST → FUNCTION TEST → API TEST → UI TEST → FIX → RETEST
```

2. Functional Testing

Check form submission, validation, Gemini request, response handling, result display and regeneration.

Tabulation

Test Type	What is checked
Functional	Features work as expected
Integration	Frontend/backend/Gemini communication
UI	Layout and readability
Error	Invalid input/API failure handling

Coding / Technical Example

```
def test_plan(plan):
    assert plan is not None
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test_plan("Sample FitBuddy plan")
```

Picture / Diagram

```
INPUT TEST → FUNCTION TEST → API TEST → UI TEST → FIX → RETEST
```

3. Integration Testing

Test communication between the frontend, backend and Gemini API. Also test behavior when the API is unavailable.

Tabulation

Test Type	What is checked
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Coding / Technical Example

```
def test_plan(plan):
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